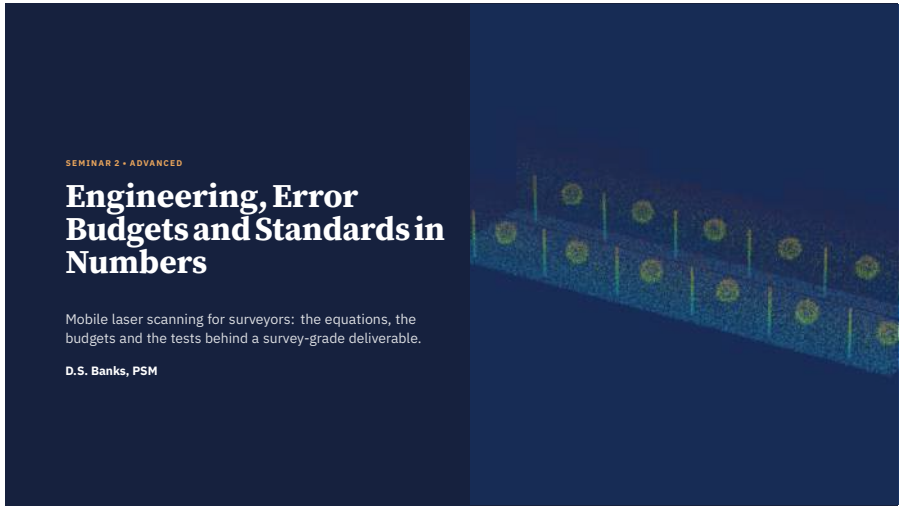
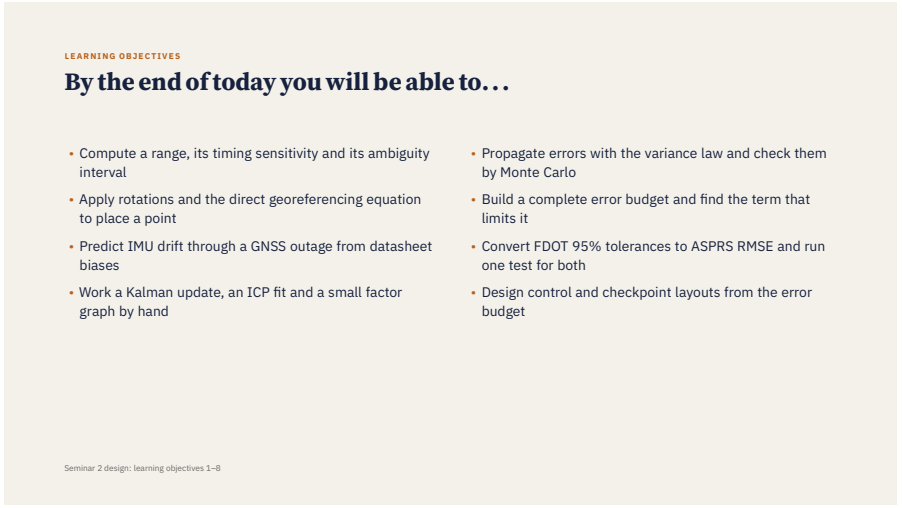


Welcome back. Seminar 1 gave the concepts with no calculations. Today is the other half of the book: the engineering track. Everyone needs a calculator and the worked-examples workbook open. Every number on these slides is worked in the Reference Book, so nobody has to copy anything down.



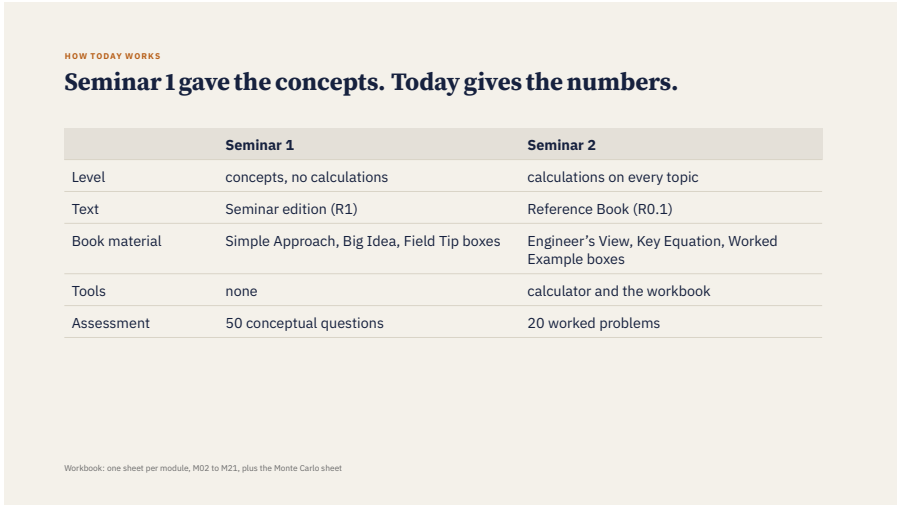
Read the objectives aloud. The problem set at the end has at least two problems from each session, so nothing in it is outside this list. Objectives 7 and 8 are what most people came for; the first six are what make them possible.

TODAY

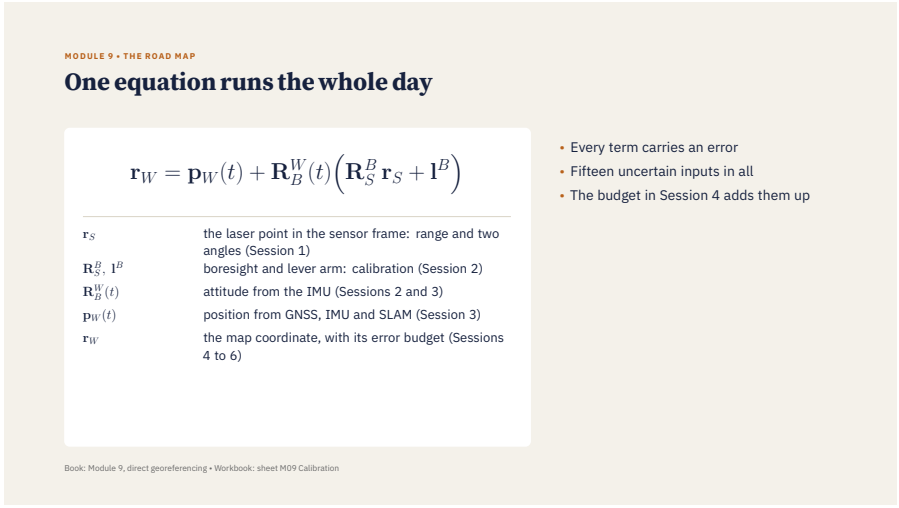
Agenda

Time	Session	Reference Book
0:00–0:15	Welcome, recap, the road map	Module 9
0:15–1:05	1. The single measurement in numbers	Modules 3, 4, 5
1:05–1:55	2. Pose, IMU and calibration math	Modules 6, 7, 8, 9
1:55–2:10	Break	
2:10–3:20	3. Estimation: how the trajectory is computed (core block)	Modules 10, 11, 12
3:20–3:50	Lunch	
3:50–4:50	4. Error propagation and error budgets	Module 14
4:50–5:35	5. Hardware in numbers	Modules 15, 16, 17
5:35–5:45	Break	
5:45–6:35	6. The standards in numbers	Modules 13, 19, 21
6:35–7:05	Problem set and evaluation	

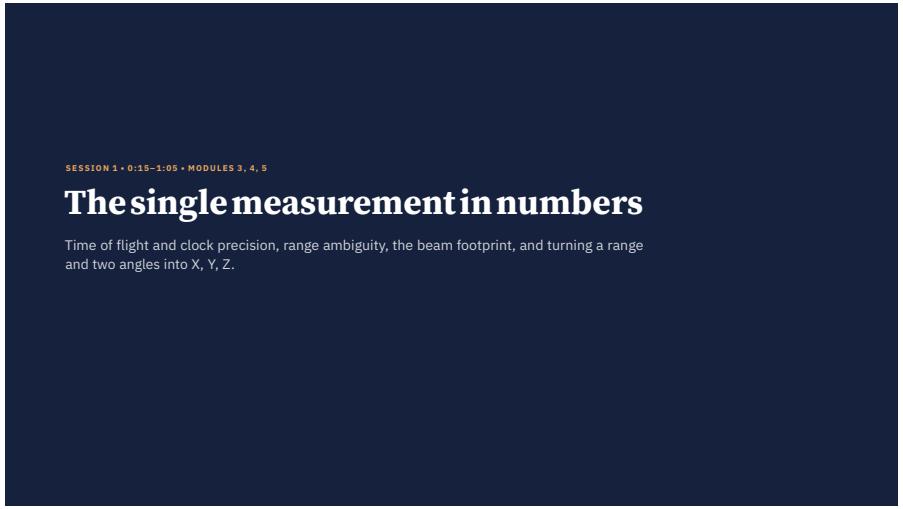
Six hours of instruction plus breaks. Confirm the contact hours the Board will credit before advertising. Session 3 is the core block at 70 minutes, as in Seminar 1. Session 6 is the one most people came for: protect its full 50 minutes even if earlier sessions run long.



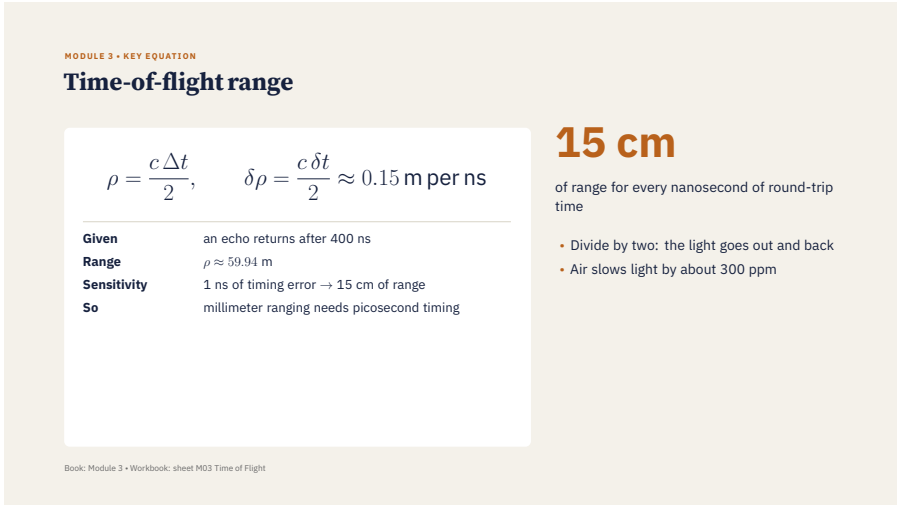
Set expectations. Every slide today shows an equation or a worked result, with its book module and workbook sheet at the foot. The workbook has live formulas: change a blue input and the answer recalculates. Encourage people to put in their own sensor's numbers as we go.



This is the direct georeferencing equation, and it is the outline for the day. Read it right to left: a laser point in the sensor frame, rotated by the boresight, shifted by the lever arm, rotated by the vehicle attitude, and added to the vehicle position. Each session fills in one term. By Session 4 we will have an uncertainty for every one of the fifteen inputs.



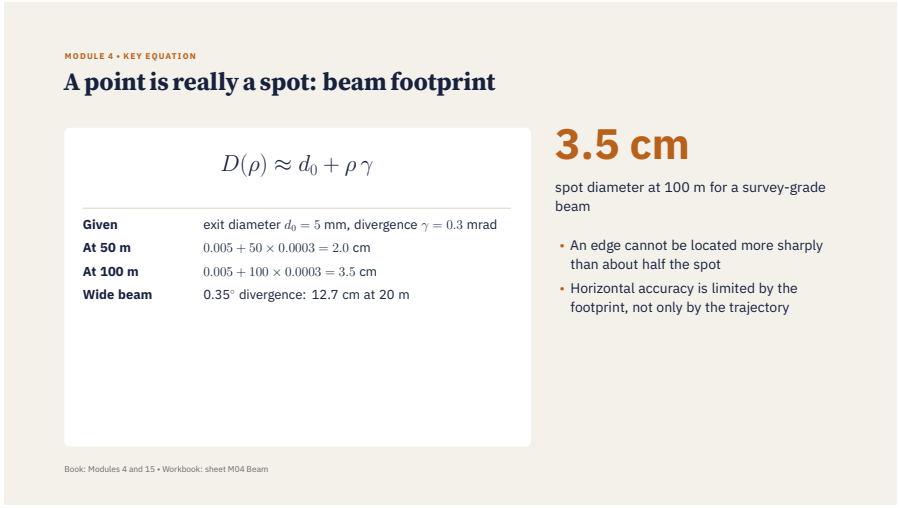
Fifty minutes. Goal: everyone can compute what one laser return is worth, in centimeters, before any trajectory or calibration is involved. This is the $\mathbf{r}_S$ term of the road-map equation.



Light covers about 30 cm per nanosecond; out and back makes it 15 cm of range. Work the 400 ns echo together: 59.94 m. The atmospheric correction is about 300 parts per million, which software applies from temperature and pressure. Hold on to that number: at 500 ft it is 4.6 cm, and it comes back in Session 5.



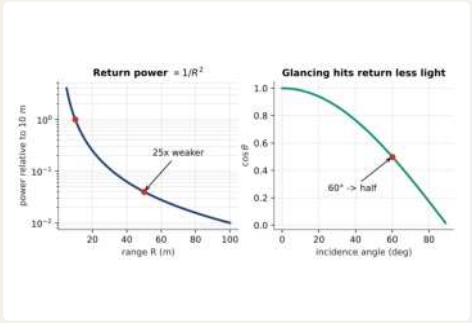
If the next pulse leaves before the last echo is home, the receiver cannot tell which pulse an echo belongs to. At 1 MHz the safe range is 149.9 m; double the rate and it halves. Survey-grade scanners fire faster than this limit and sort the echoes into zones in processing. A wrong zone assignment produces ghost layers tens of meters off, which is a blunder you look for, not noise you average.



The footprint is the exit diameter plus range times divergence. A survey-grade beam is 2 to 3.5 cm across at mapping ranges. An automotive-style sensor with 0.35 degrees of divergence is 12.7 cm across at 20 m. This matters in Session 6: FDOT Type A asks for 0.06 ft horizontally, which is 1.8 cm, tighter than the spot itself.

MODULE 4 • THE RANGE EQUATION

Echo strength sets the usable range



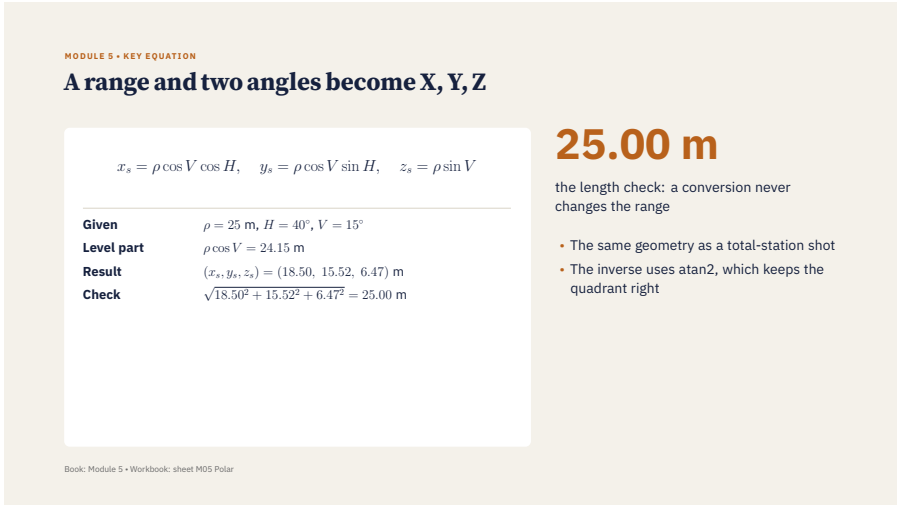
25 m

usable on fresh asphalt for a sensor rated 35 m on a 10% target

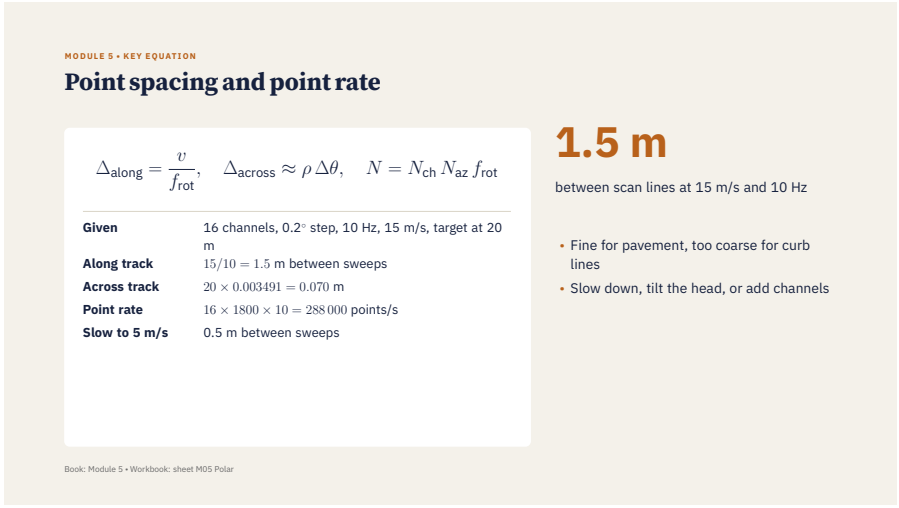
- Received power falls as $1/R^2$ and with $\cos\theta_i$
- Range scales with $\sqrt{\text{reflectivity}}$: $35 \times \sqrt{5/10} = 25$ m
- Plan the swath on the darkest surface you must map

Book: Module 4 • Workbook: sheet M04 Beam

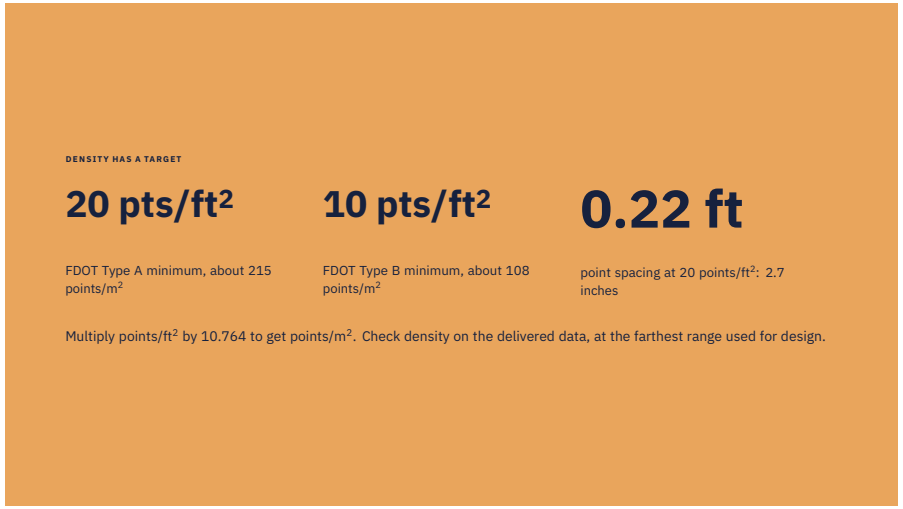
Received power falls with the square of range and with the cosine of the incidence angle. For the same detection threshold, maximum range grows with the square root of reflectivity. A sensor rated 35 m on a 10 percent target reaches about 25 m on 5 percent fresh asphalt. Weak echoes also have noisier ranges, so the far edge of the road is the noisiest part of a mobile scan.



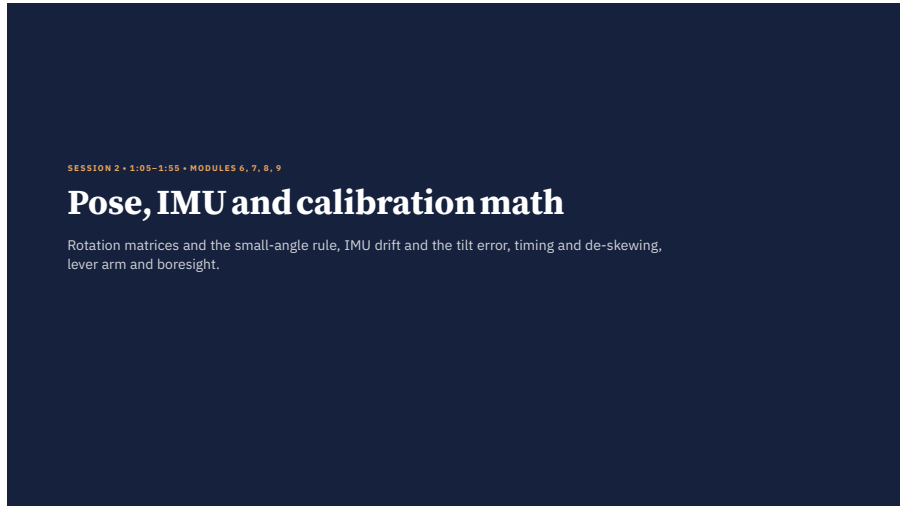
This is a total-station reduction in the sensor's own frame: tilt first, then turn. Work the example: 25 m at 40 degrees horizontal and 15 degrees vertical gives 18.50, 15.52, 6.47. Always check the length. This is the $\mathbf{r}_S$ term of the road map, and its three inputs, range and two angles, are the first three lines of the error budget.



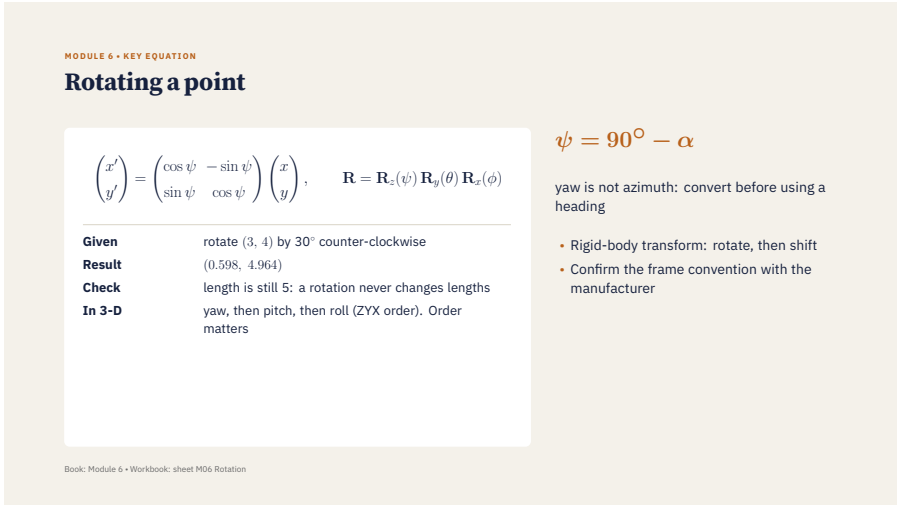
Speed over rotation rate gives the along-track spacing. Sixteen channels at 15 m/s leave 1.5 m between sweeps of each channel. Have everyone change the speed in the workbook and watch the spacing move. Tilting the head closes the gaps because several channels then cross the same strip of road.



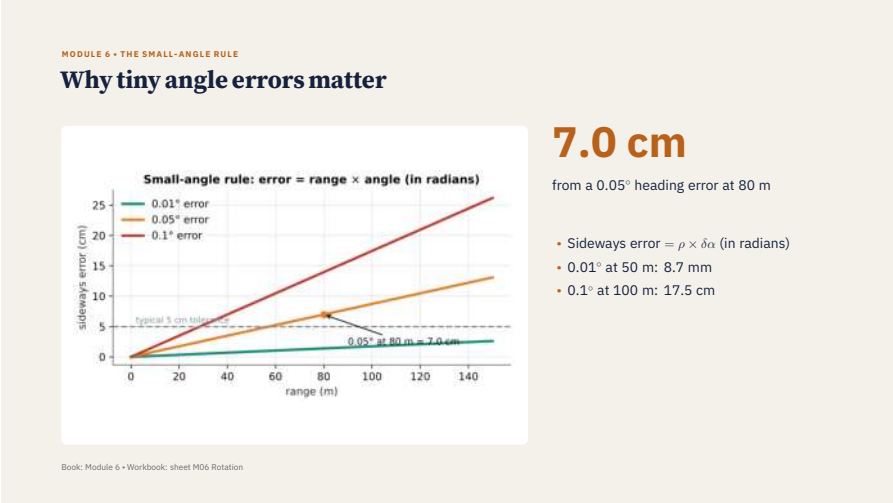
This is the takeaway of Session 1. FDOT states density per square foot. At the Type A minimum of 20 points per square foot, neighboring points are 0.22 ft apart, nearly four times coarser than the 0.06 ft horizontal tolerance. Keep that in mind for Session 6. Transition: we know what one return is worth in the sensor frame. Next, where the sensor is and which way it faces.



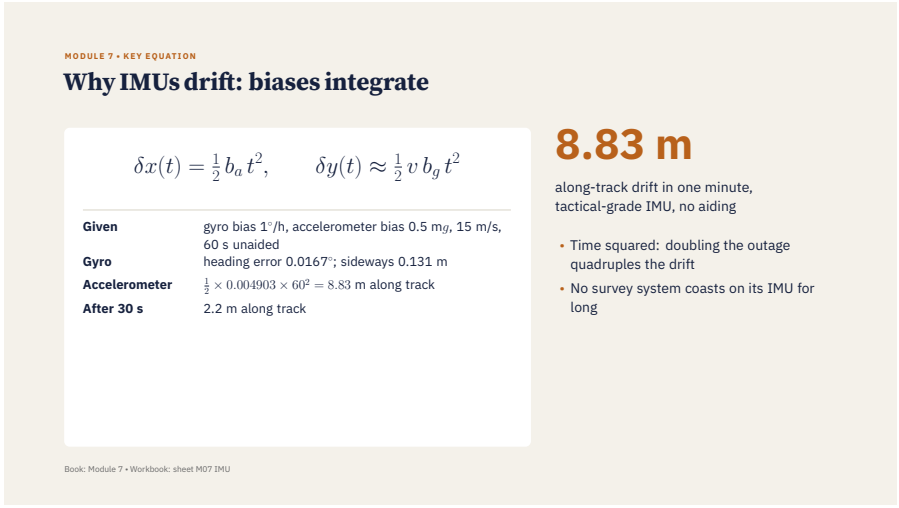
Fifty minutes. Everything in this session is an angle or a time multiplied by something: angle error times range, timing error times speed, timing error times rotation rate times range. Have participants compute all three for their own system.



Work the two-dimensional rotation by hand once: 3, 4 rotated 30 degrees becomes 0.598, 4.964, and the length is still 5. In three dimensions the order is yaw, pitch, roll, and swapping the order gives a different answer. Remind them that survey azimuth is clockwise from north and robotics yaw is counter-clockwise from east. A frame or sign mistake here is a blunder, not an error budget term.



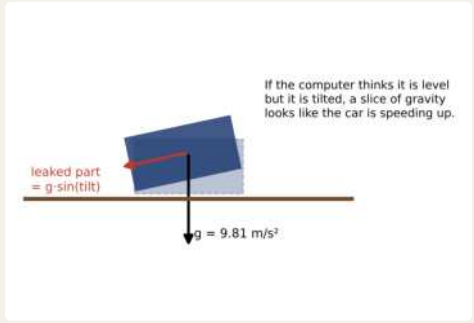
This is the most used rule of the day: an angle error in radians times the range. One hundredth of a degree is 1.745 times ten to the minus four radians. At 80 m, 0.05 degrees is 7.0 cm. Angle errors grow with distance, so a system that is excellent at 15 m can be unacceptable at 80 m. This is why far objects are where you look for calibration problems.



The IMU integrates its readings, so a constant bias grows with time squared. A tactical-grade unit with half a milli-g of accelerometer bias drifts almost 9 m along track in one minute with no aiding. After 30 seconds it is 2.2 m. This single calculation is why control goes inside every GNSS outage, which is where Session 6 ends up.

MODULE 7 • THE HIDDEN KILLER

A tilt error leaks gravity into acceleration



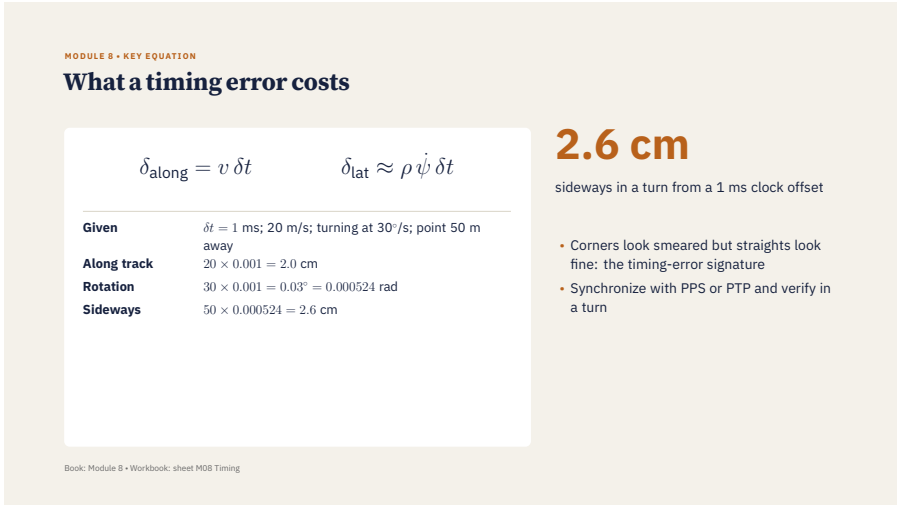
0.86 m

after 10 s from a 0.1° tilt error

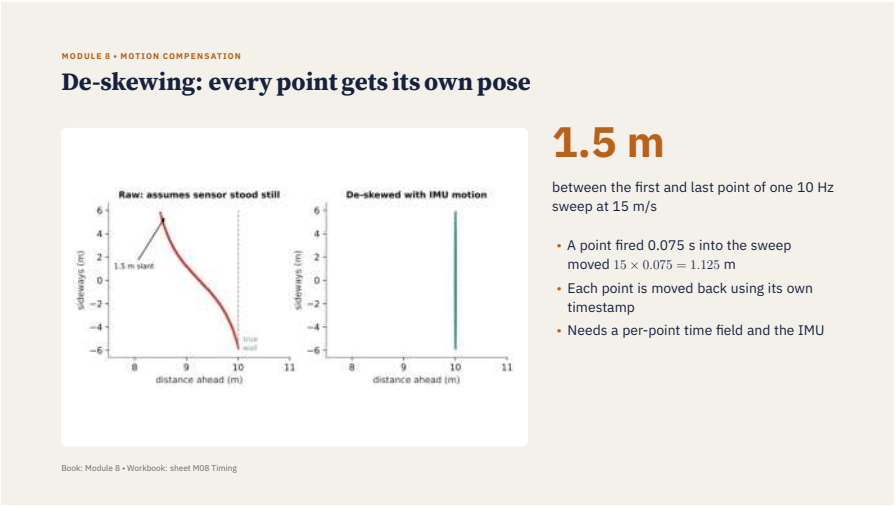
- Leaked acceleration $= g \sin 0.1^\circ = 0.0171$ m/s²
- That is 1.7 mg: worse than the accelerometer bias
- Attitude must be known very well, or gravity becomes the biggest error

Book: Module 7 • Workbook: sheet M07 IMU

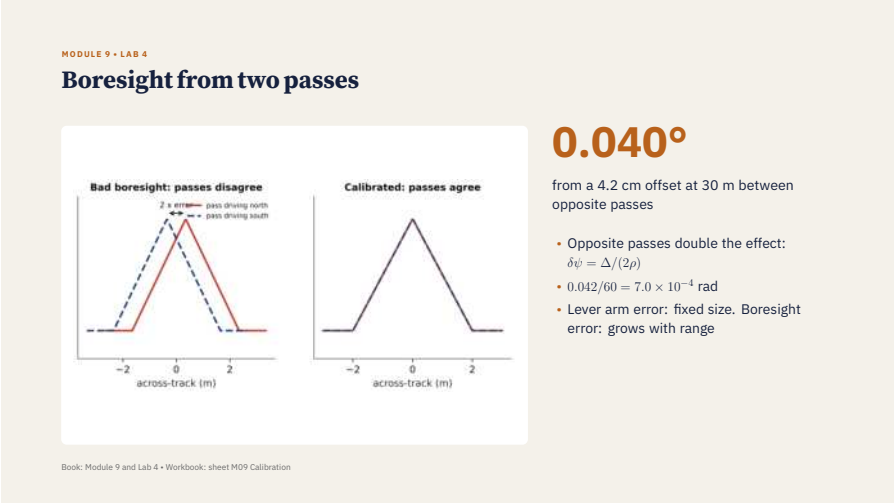
This is the idea most people have not met before. If the computer thinks the unit is level and it is tilted by a tenth of a degree, a slice of gravity looks like the vehicle speeding up. That is 1.7 milli-g, three times the accelerometer bias on the last slide, and 0.86 m after ten seconds. Good filters keep estimating roll and pitch from the gravity direction whenever the vehicle is not accelerating.



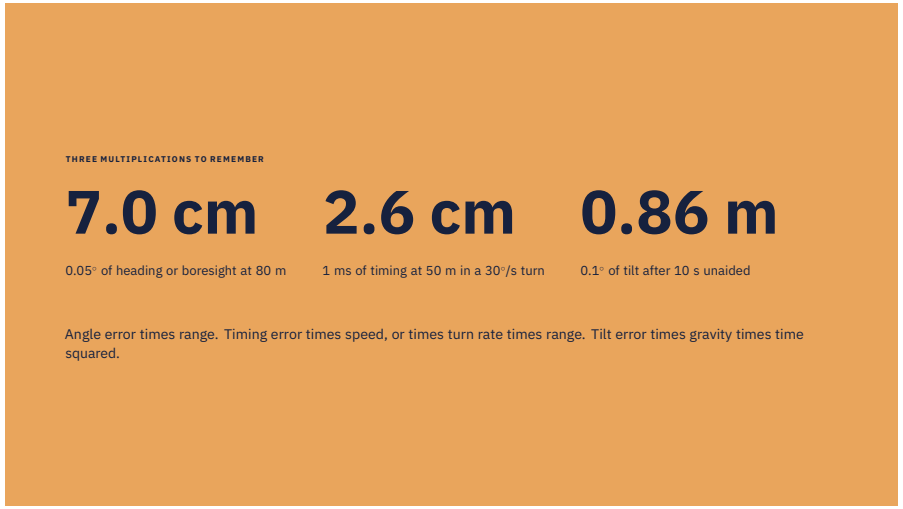
Two terms. Along track it is speed times the offset: 2 cm at 20 m/s for one millisecond. Sideways it is range times turn rate times the offset: 2.6 cm at 50 m in a 30 degree per second turn. Because the second term needs a turn, the signature of a timing error is smeared corners and clean straights. That is how you diagnose it in the cloud.



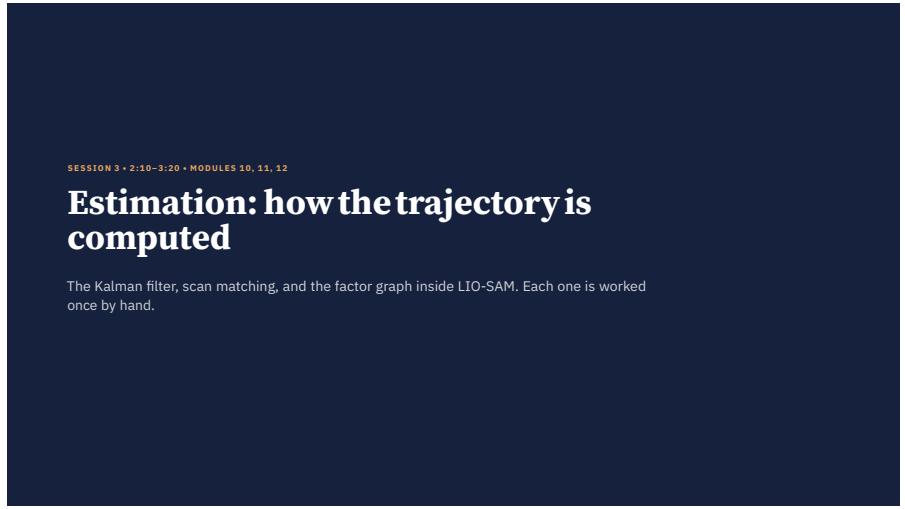
A spinning scanner takes a tenth of a second per revolution, and at 15 m/s the vehicle moves 1.5 m in that time. Without de-skewing, the points of a single sweep disagree by up to 1.5 m. De-skewing gives each point the pose for its own timestamp. Before processing, confirm the point cloud has a per-point time field; without it, nothing downstream can fix this.



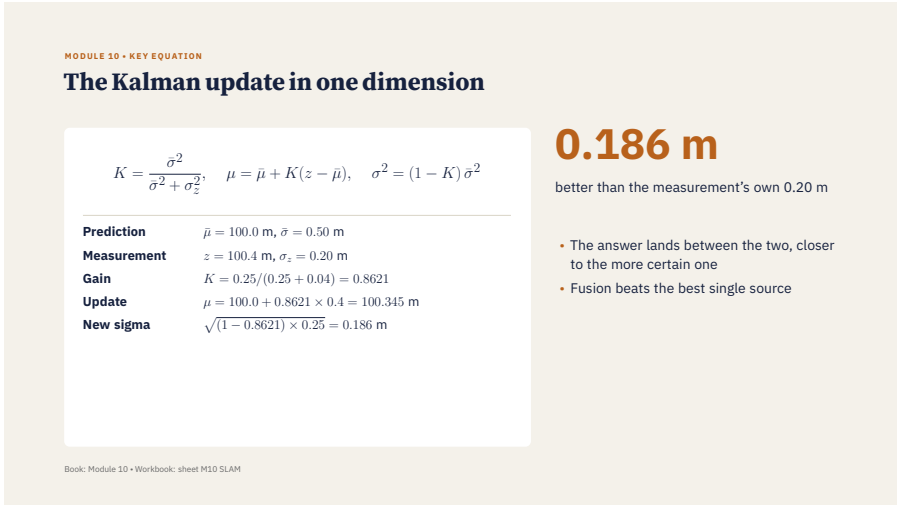
This is Lab 4, done together. Drive past the same building corner in both directions. A yaw boresight error shifts the feature forward on one pass and backward on the other, so the passes differ by twice range times the error. A 4.2 cm offset at 30 m is 0.040 degrees. Compare with the lever arm: a 7 mm lever-arm error is 7 mm everywhere, while a 0.05 degree boresight error is 7 cm at 80 m.



The takeaway of Session 2. Ask everyone to compute the first two for their own system and typical range before the break. Transition: we now have every term of the road-map equation except the big one, the trajectory. After the break: how the trajectory is actually computed.



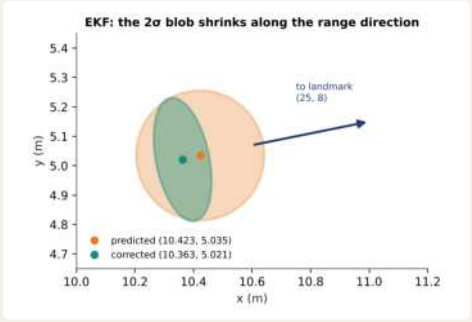
Seventy minutes, the core block. Work each method once by hand before showing any software. The Kalman update and the four-pose factor graph both fit on one slide and one workbook row. Tie everything to least-squares adjustment, which this audience already knows.



Work this one line by line. A prediction good to half a meter and a measurement good to 20 cm combine into an estimate good to 18.6 cm, better than either. The gain K is just the weight: how much to trust the measurement. This is a weighted mean, exactly what a least-squares adjustment does with two observations of the same distance.

MODULE 10 • THE EXTENDED KALMAN FILTER

Moving grows uncertainty; a landmark shrinks it



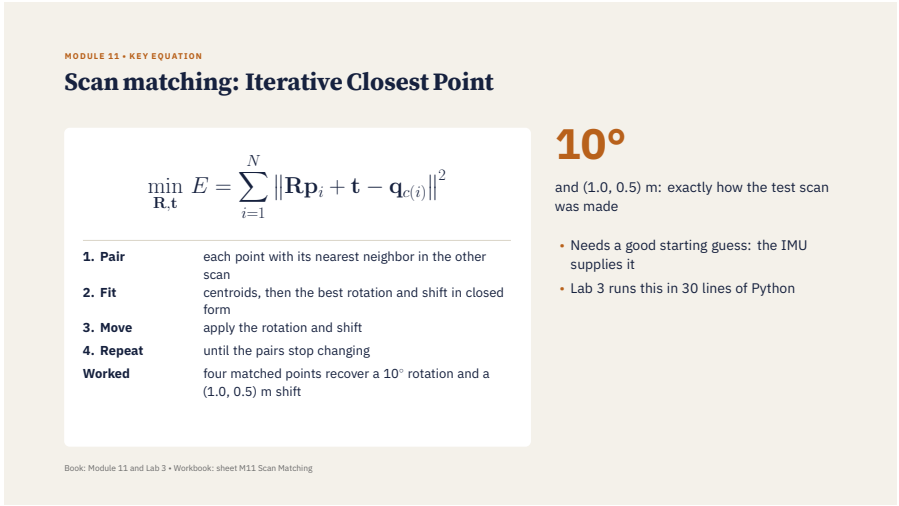
11.0 to 4.9 cm

in x, after one range and bearing to a pole

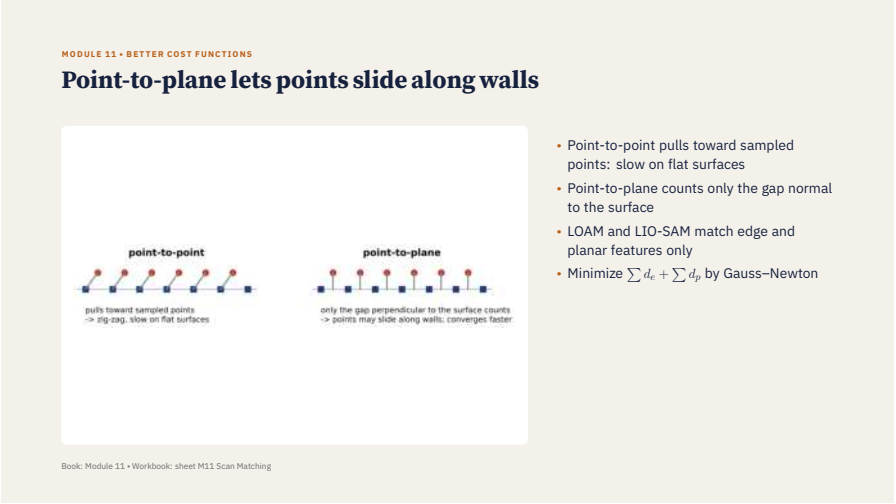
- Predict: $\bar{\mathrm{P}} = \mathrm{G}\,\mathrm{P}\,\mathrm{G}^{\mathsf{T}} + \mathrm{Q}$
- y barely improved: the pole is almost straight ahead
- A range only constrains the direction it points

Book: Module 10 • Workbook: sheet M10 SLAM

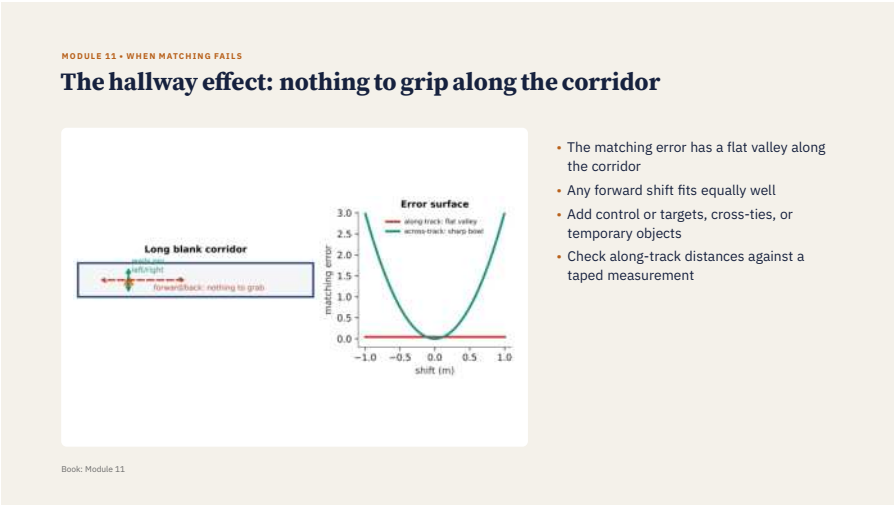
The extended filter is the same idea with matrices. Each prediction inflates the uncertainty ellipse; one good landmark observation collapses it. Seeing a pole almost straight ahead tightens x from 11.0 to 4.9 cm but hardly helps y, because a range measurement only constrains the direction it points. Geometry of the observations matters, exactly as in a resection.



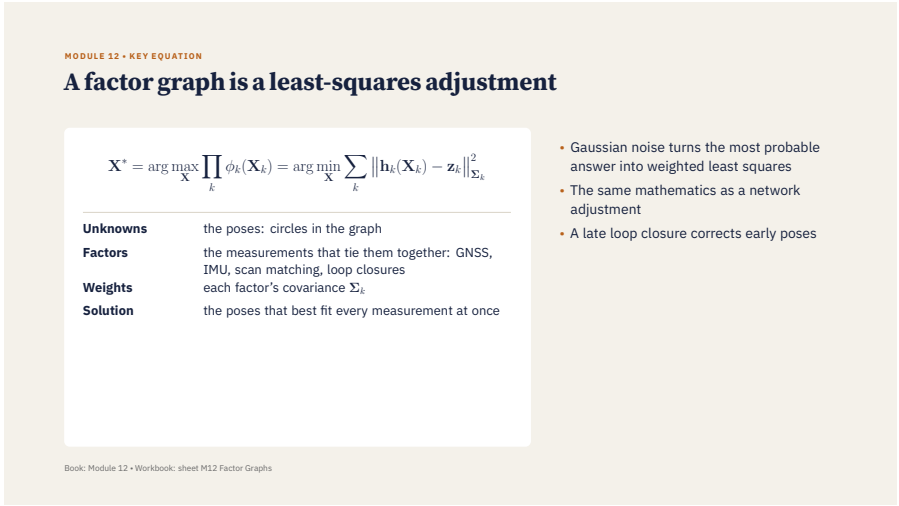
ICP is a two-point resection done with thousands of points. Pair, fit, move, repeat. With correct pairs the best fit is a closed-form calculation; the worked example recovers the 10 degree rotation and the shift exactly. In real data the pairs are guesses, so the steps repeat. A poor starting guess can converge to the wrong answer, which is why the IMU prediction comes first. Run Lab 3 here if time allows.



Two scans never sample the same points, so pulling a point toward its nearest neighbor is slightly wrong. Point-to-plane penalizes only the distance perpendicular to the surface and converges faster. LIO-SAM goes further: it keeps only sharp edge points and flat planar points, and minimizes point-to-line and point-to-plane distances.



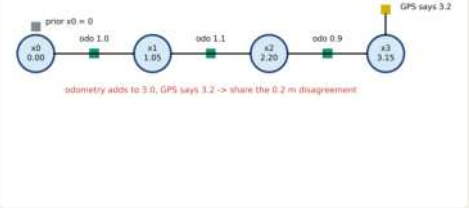
In a featureless corridor, or beside a long plain wall, the walls pin the scan left and right but nothing constrains forward and back. The error surface is a flat valley. The same thing happens on an open road with no vertical features. This is the scene-geometry weakness that the trajectory's reported uncertainty should reveal, and it is one reason control spacing cannot be a fixed number of feet.



This is the one equation to recognize in the LIO-SAM paper. Under Gaussian noise, the most probable trajectory is the one that minimizes the weighted sum of squared residuals. Every surveyor in the room has run this: it is a least-squares network adjustment, where the stations are poses and the observations are factors. The weights are the covariances, which is why a wrong weight does damage.

MODULE 12 • WORKED BY HAND

Four poses, three odometry legs, one GPS fix



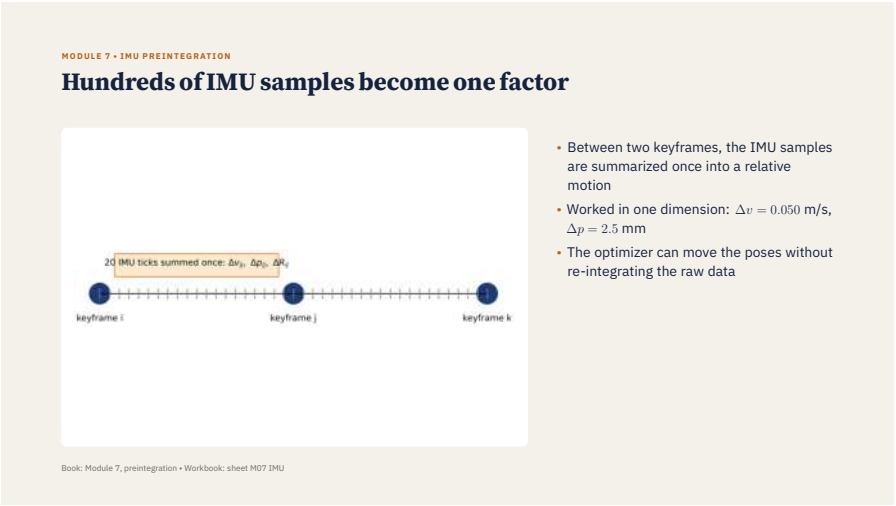
0.20 m

misclosure, shared equally by four equally weighted measurements

- Odometry: 1.0, 1.1, 0.9 m. GPS says 3.2 m
- Solution: $x_1 = 1.05$, $x_2 = 2.20$, $x_3 = 3.15$ m
- A traverse adjustment, done by a factor graph

Book: Module 12 • Workbook: sheet M12 Factor Graphs

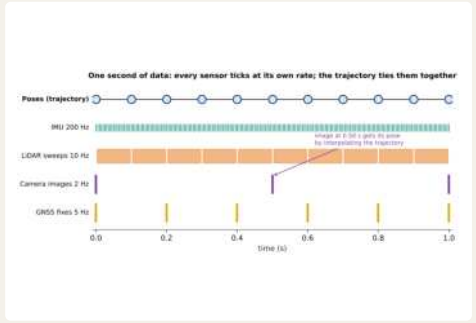
Solve this one on the board. Three odometry legs sum to 3.0 m and GPS says 3.2 m. With equal weights the 0.20 m misclosure is shared equally: each of the four measurements takes 5 cm. The answer is 1.05, 2.20, 3.15. That is the compass rule. Then change the GPS weight in the workbook and watch the solution lean toward the more trusted measurement.



The optimizer adjusts the poses many times. Re-integrating thousands of raw IMU samples every time would be too slow, so the samples between two keyframes are summarized once into a change in velocity, position and rotation. Ten numbers become two in the worked example. That summary is the IMU factor between consecutive poses.

MODULE 12 • LIO-SAM

One second of data, one common thread



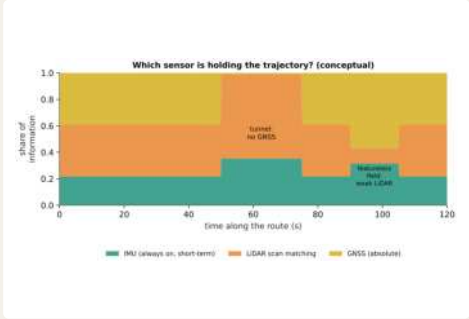
- IMU: 200 ticks. LiDAR: 10 sweeps. GNSS: 5 fixes. Camera: 2 images
- State: $\mathbf{x} = [\mathbf{R},\ \mathbf{p},\ \mathbf{v},\ \mathbf{b}]$
- The trajectory gives every measurement its pose
- Only keyframes enter the graph

Book: Module 12 • Workbook: sheet M12 Factor Graphs

Put the pieces together. In one second the IMU ticks 200 times, the scanner completes 10 sweeps, GNSS gives 5 fixes and the camera takes 2 images. The trajectory is the common thread: every laser pulse and every image gets its pose by interpolating it. The state carries rotation, position, velocity and the IMU biases, so the filter keeps re-estimating the biases from Session 2.

MODULE 12 • WHO CARRIES THE TRAJECTORY

The graph leans on whichever sensor has information



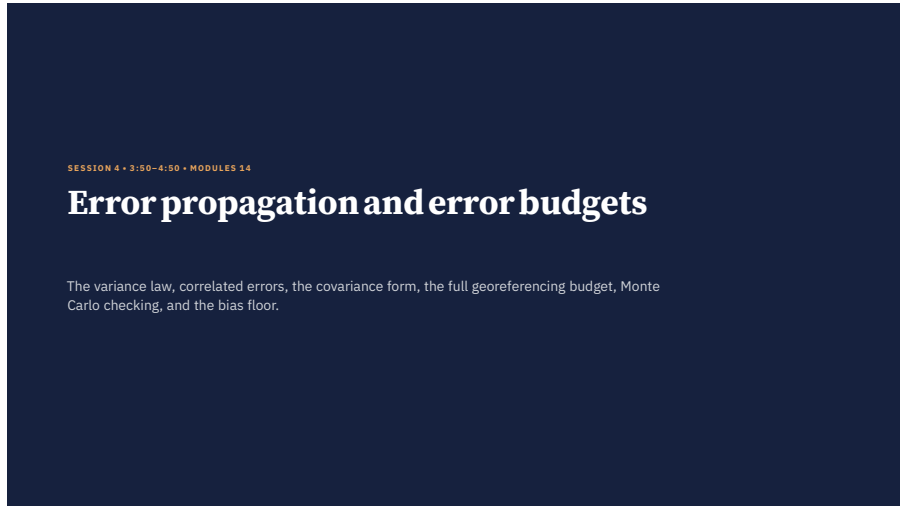
- Open sky: GNSS and IMU
- Tunnel: LiDAR matching and IMU
- Featureless field: GNSS and IMU
- Inspect the trajectory's standard deviation before trusting any part of the cloud

Book: Module 12

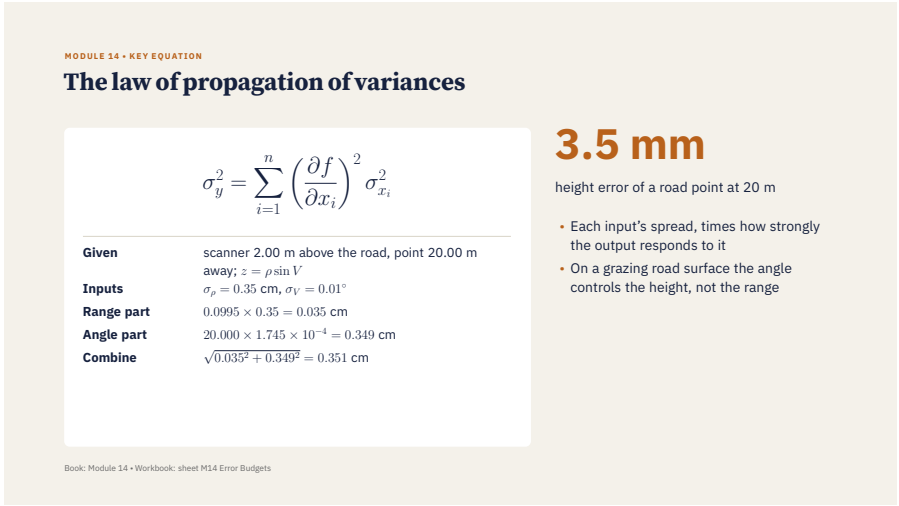
This figure is the bridge to Session 6. The sensors have different jobs. In the tunnel there is no GNSS, so scan matching and the IMU carry the solution. In the featureless field, matching is weak, so GNSS and the IMU carry it. Where two of the three are weak at once, the trajectory is weak, and that is where control and checkpoints belong.



The takeaway of the core block. SLAM is not magic and it is not limited by its worst sensor: properly weighted, it does better than its best one. What limits it is whatever the weights do not describe: a wrong GNSS fix that claims a centimeter, a timing offset, a boresight error. Those are systematic, and they are the subject of the afternoon. Lunch.



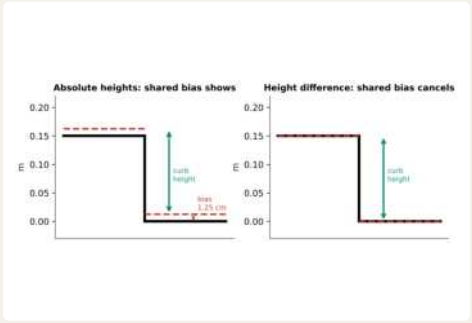
Sixty minutes, and the hardest hour. Use the variance law on one output first, then show that the matrix form is the same idea for three outputs at once. The Monte Carlo sheet lets participants confirm the formula without having to trust it.



The slope of the model converts input spread into output spread. For the height of a road point 20 m away, the range error barely matters because the beam is nearly horizontal: 0.035 cm. The vertical angle error contributes 0.349 cm, ten times more. On a wall straight ahead it is the other way round. Knowing which input controls which output is the whole skill.

MODULE 14 • CORRELATED ERRORS

A shared bias cancels in a difference



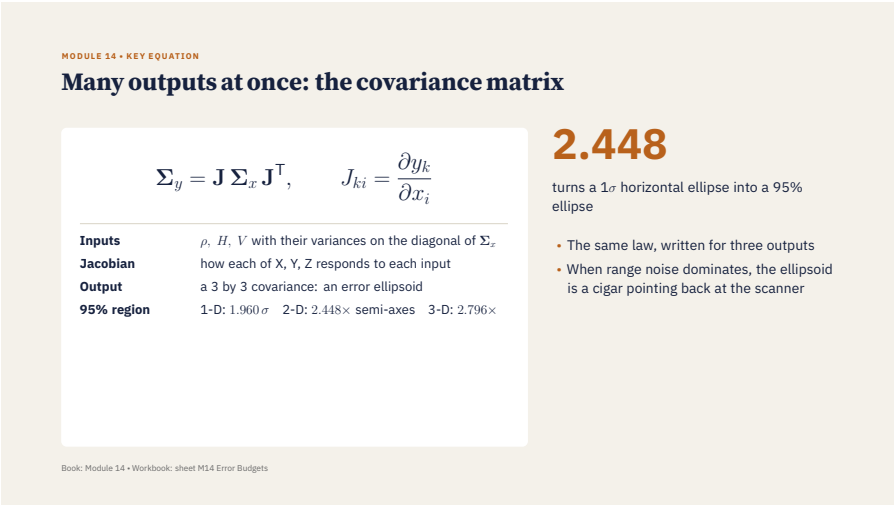
3.5 mm

curb height, though each absolute height is good only to about 1.3 cm

- $\sigma_y^2 = \sum a_i^2 \sigma_i^2 + 2 \sum a_i a_j \sigma_{ij}$
- Same sweep, same 1.25 cm range bias on both points
- Relative measurements beat absolute ones

Book: Module 14 • Workbook: sheet M14 Error Budgets

When two measurements share an error, the covariance term matters. A curb top and gutter measured in the same sweep both carry the same 1.25 cm range bias. In the height difference the bias cancels, leaving only the random part: 3.5 mm. This is why a point cloud's relative accuracy is so much better than its absolute accuracy, and why a crisp cloud says nothing about where it is.



This is the same law for several outputs at once. The Jacobian holds the sensitivities; multiply through and you get a covariance matrix for X, Y and Z. Its off-diagonal terms are real: when the beam runs diagonally, x and y errors are correlated. The 95 percent factors are different in one, two and three dimensions: 1.960, 2.448, 2.796. The NSSDA horizontal factor of 1.7308 that FDOT uses is the circular version of the two-dimensional case.

MODULE 14 • WORKED EXAMPLE

Full error budget of a backpack point at 20 m (1σ, cm)

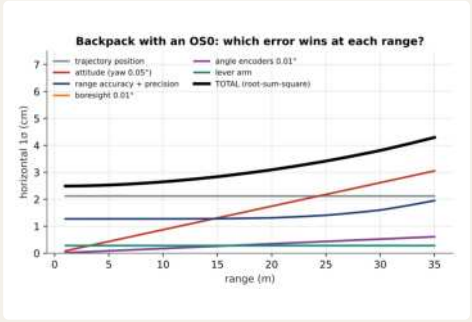
Term	Horizontal	Vertical
Trajectory position (SLAM + control)	2.12	2.00
Attitude: yaw 0.05°, roll and pitch 0.02°	1.75	0.70
Boresight 0.01° per axis	0.35	0.35
Lever arm 2 mm per axis	0.28	0.20
Range: accuracy 1.25, precision 0.40	1.31	—
Angle encoders 0.01°	0.35	0.35
Total (root-sum-square)	**3.10**	**2.18**

Book: Module 14 • Workbook: sheet M14 Error Budgets

Here is the road-map equation with a number on every term. Square each term, add, take the root: 3.10 cm horizontal and 2.18 cm vertical. The trajectory and the heading are the two biggest terms; the laser is third. Have everyone replace these inputs with their own system's values in the workbook. The exercise is to find which term to attack first.

MODULE 14 • THE BUDGET AGAINST RANGE

Fixed terms dominate close in; angle terms take over far out



18 m

where this system stops meeting a 3 cm horizontal specification

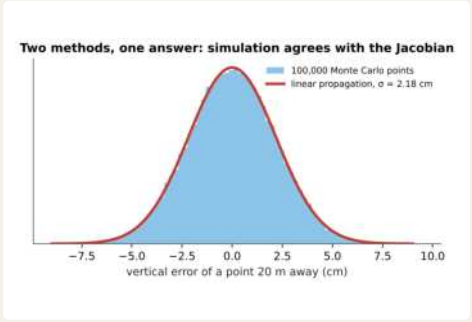
- Use the curve to set the maximum useful range
- Beyond it: filter points by range, or improve heading

Book: Module 14 • Workbook: sheet M14 Error Budgets

Plot the budget against range and it tells you the maximum useful range. Close in, the trajectory and lever arm dominate and do not change with distance. Far out, every angle term grows in proportion to range. For this backpack a 3 cm horizontal specification is met only out to about 18 m. Beyond that you either exclude the points from the design surface or improve the heading.

MODULE 14 • CHECKING THE FORMULA

Monte Carlo: simulate it 100,000 times



under 1%

difference between simulation and the linear formula

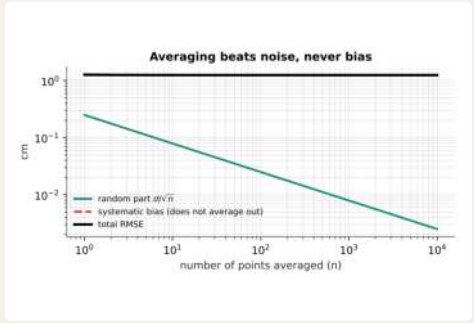
- Horizontal: 3.09 cm simulated, 3.10 cm by formula
- Vertical: 2.18 cm both ways
- No derivatives needed; handles any distribution

Book: Module 14 • Workbook: sheet MC 20 m

If the algebra feels like an act of faith, check it. Draw random values for all fifteen inputs, run the full equation with rotation matrices, repeat 100,000 times, and measure the spread. Horizontal 3.09 against 3.10; vertical 2.18 against 2.18. Open the Monte Carlo sheet and press recalculate. For angles of a few hundredths of a degree the linear method is essentially exact.

MODULE 14 • AVERAGING

Averaging shrinks noise but not bias



1.25 cm

the floor: 400 returns on a target average the noise away, not the bias

- RMSE$_p = \sqrt{b^2 + \sigma^2/n}$
- Random part: $0.25/\sqrt{400} = 0.0125$ cm
- Only calibration against known distances removes a bias

Book: Module 14 • Workbook: sheet M14 Error Budgets

Fit a plane to 400 returns on a flat target. The noise drops by a factor of 20, to 0.0125 cm, and the accuracy barely changes, because the 1.25 cm range bias is still there. Averaging cannot go below the systematic error. Hold this thought for Session 6: FDOT Type A needs 0.93 cm, which is already below this sensor's bias floor.

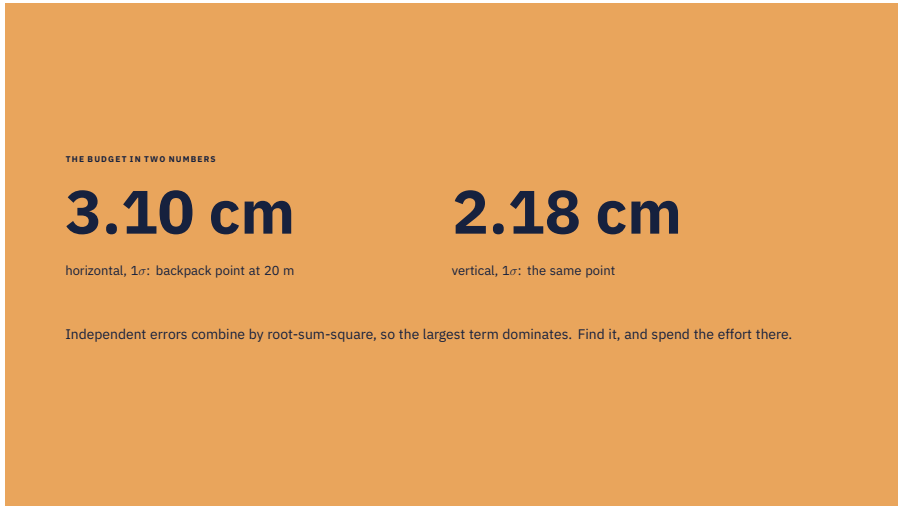
MODULE 14 • WORKED EXAMPLE

A root-sum-square budget for a 5 cm job (1σ, cm)

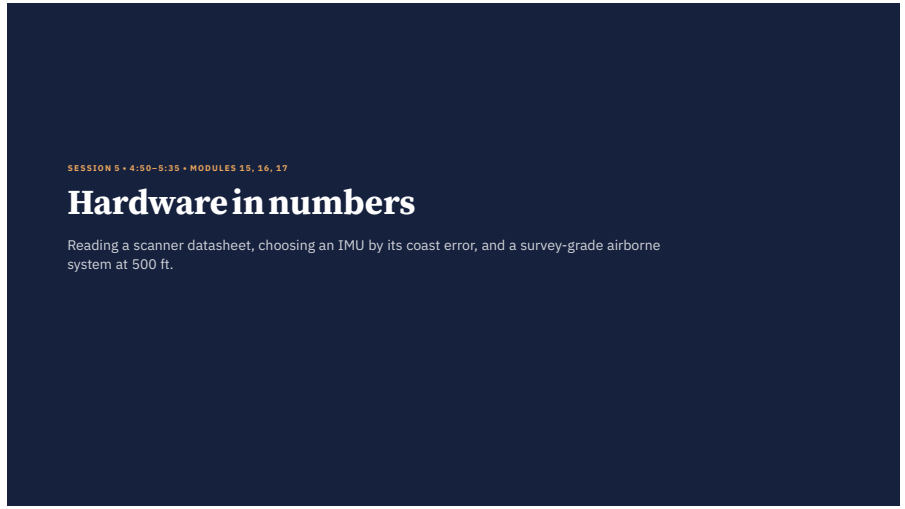
Term	As planned	Featureless stretch
Range and angle	2.0	2.0
Lever arm and boresight	1.5	1.5
Trajectory	3.0	**5.0**
Timing	0.5	0.5
Control and georeferencing	1.5	1.5
Total	**4.21: passes**	**5.81: fails**

Book: Module 14 • Workbook: sheet M14 Error Budgets

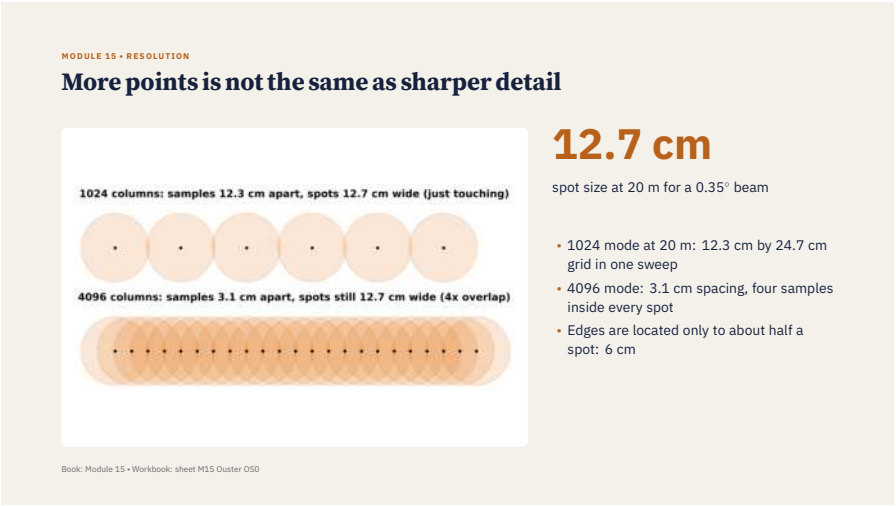
A budget is a planning tool. As planned, the total is 4.21 cm, inside 5. In a long featureless stretch the trajectory term rises to 5.0 and the total becomes 5.81: the job fails. The fix is not a better laser. It is more control in the weak stretch, a slower pass, or a return pass for loop closures. The largest term decides where the money goes.



The takeaway of Session 4. Terms of 1, 1 and 3 cm total 3.3 cm: improving the small ones barely helps. In almost every mobile LiDAR budget the trajectory is the largest term, which is why the rest of the day is about the trajectory, the control that corrects it and the tests that prove it.



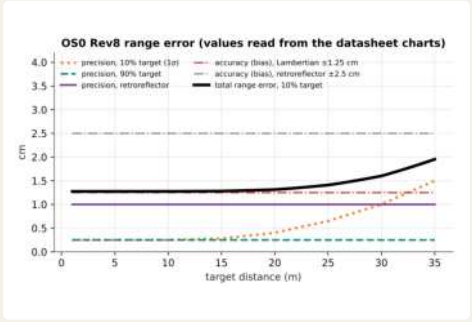
Forty-five minutes. Use the case studies as budget inputs, not as product reviews. The point of each calculation is the decision it supports.



A datasheet’s angular resolution is not its real resolution. At 20 m this sensor’s spot is 12.7 cm wide. In its 1024 mode the samples sit edge to edge. In the 4096 mode there are four samples inside every spot: more points, but each one still averages the same 12.7 cm of wall. The beam footprint is the real resolution limit.

MODULE 15 • RANGE ERROR

Total range error from a datasheet



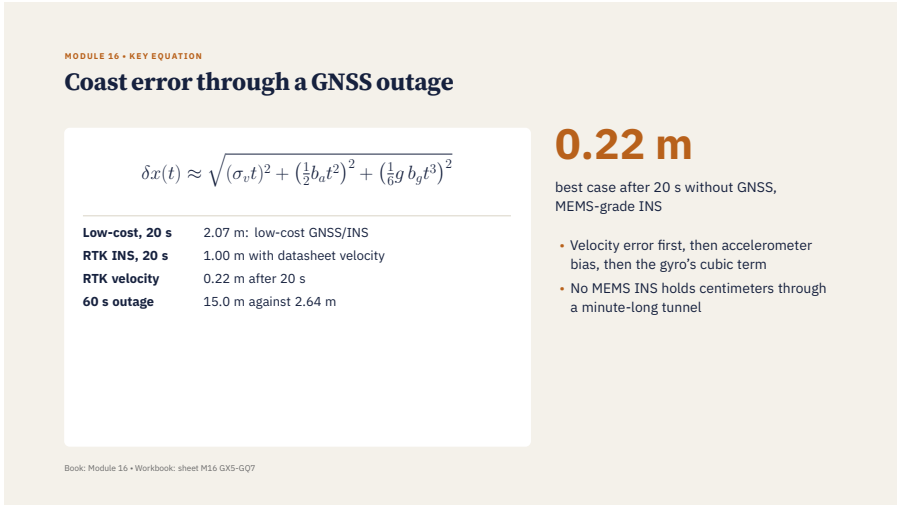
1.25 cm

range bias: it dominates out to about 25 m

- $\sigma_\rho \approx \sqrt{b^2 + \sigma_\beta^2 + \sigma_q^2}$
- Dark asphalt: 1.27 cm at 10 m, 1.31 at 20 m, 1.60 at 30 m, 1.95 at 35 m
- Retroreflective targets are worse: use matte targets for height

Book: Module 15 • Workbook: sheet M15 Ouster OS0

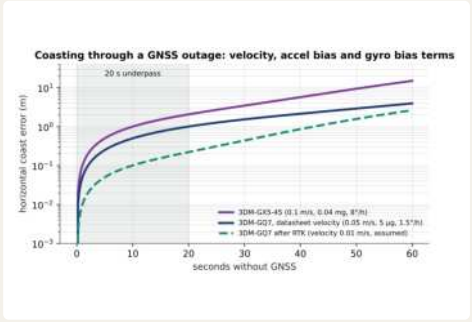
Resolution, precision and accuracy are three different lines on a datasheet. Combine bias, precision and quantization by root-sum-square. On dark asphalt the total is 1.27 cm at 10 m and 1.95 cm at the 35 m limit. The 1.25 cm bias dominates most of the way. Note the trap: the datasheet shows a larger bias on retroreflectors, so retro tape is for horizontal identification, not for absolute height checkpoints.



This is the equation behind the FDOT Handbook's Note 4. Three terms: velocity error grows with time, accelerometer bias with time squared, gyro bias with time cubed. For a MEMS-grade INS the best case after 20 seconds is 22 cm; after a minute it is meters. Put your own IMU's datasheet values in the workbook. The answer tells you how long an outage that unit can carry for a given tolerance.

MODULE 16 • THE DECISION

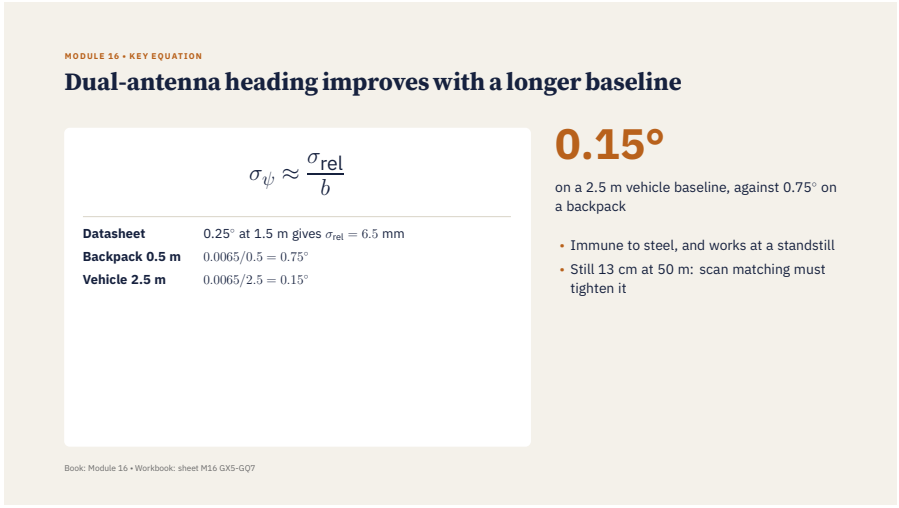
How long an outage can this IMU carry?



- Read the curve at your tolerance, not at the brochure number
- A 300 m underpass at 13 m/s is a 23 s outage
- Scan matching and control inside the outage do the rest
- State the IMU grade in the scope

Book: Module 16 • Workbook: sheet M16 GX5-GQ7

Turn the equation into a decision. Find your tolerance on the vertical axis and read across to the outage length. For centimeter work a MEMS unit carries only a few seconds unaided. A 300 m underpass at survey speed is 23 seconds. So an outage is never bridged by the IMU alone: scan matching carries it and control inside the outage pins it. This is the capstone project, and we test it against FDOT in Session 6.



Heading from two antennas is the relative position precision divided by the antenna separation. On a short backpack baseline it is no better than a magnetometer. On a vehicle roof it is 0.15 degrees. But apply the small-angle rule: 0.15 degrees at 50 m is still 13 cm. GNSS heading starts the solution; scan matching inside SLAM is what tightens it toward a few hundredths of a degree.

MODULE 17 • SURVEY-GRADE SYSTEM AT 500 FT

Where the error comes from (1σ, cm, nadir return)

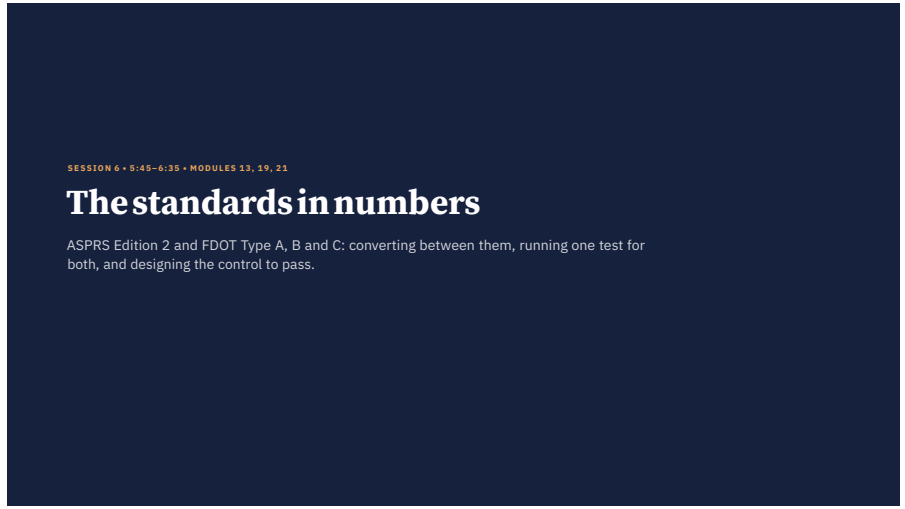
Term	Horizontal	Vertical
GNSS position	2.5	3.75
Roll and pitch 0.005°	1.88	about 0
Boresight 0.005° per axis	1.88	—
Laser range	—	2.5
Lever arm 5 mm	0.71	0.5
Total	**3.7 (0.12 ft)**	**4.5 (0.15 ft)**
With control and strip adjustment	**3.1**	**3.1**

Book: Modules 14 and 17 • Workbook: sheet M17 RIEGL

The same budget for a survey-grade airborne scanner at 500 ft. The laser's own range error is 2.5 cm. The largest term is GNSS, in both columns. Attitude of five thousandths of a degree still moves a point 1.3 cm at this range. With ground control and a strip adjustment the totals come down to about 3.1 cm. With a scanner this good, the laser is no longer the weak link.



The takeaway of Session 5. As the sensors improve, the random errors shrink and what is left is systematic: the GNSS floor, calibration, the atmospheric correction, the reference. Those are the surveyor's part. Short break, then the standards.



Fifty minutes, and the session most participants came for. Keep the conversion table on screen as long as possible. The Recommendations document in the Supplementary Information has everything in this session written out.

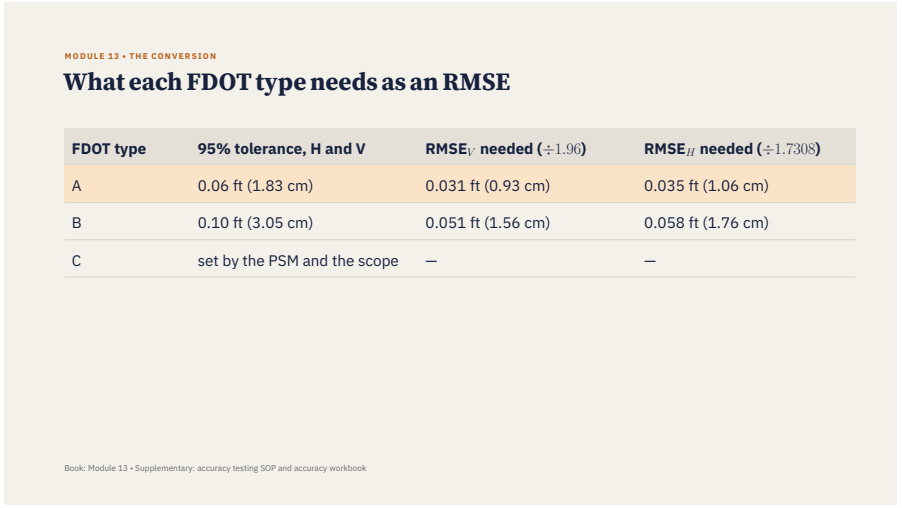
MODULES 13 AND 19 • TWO STANDARDS

ASPRS Edition 2 and FDOT test accuracy differently

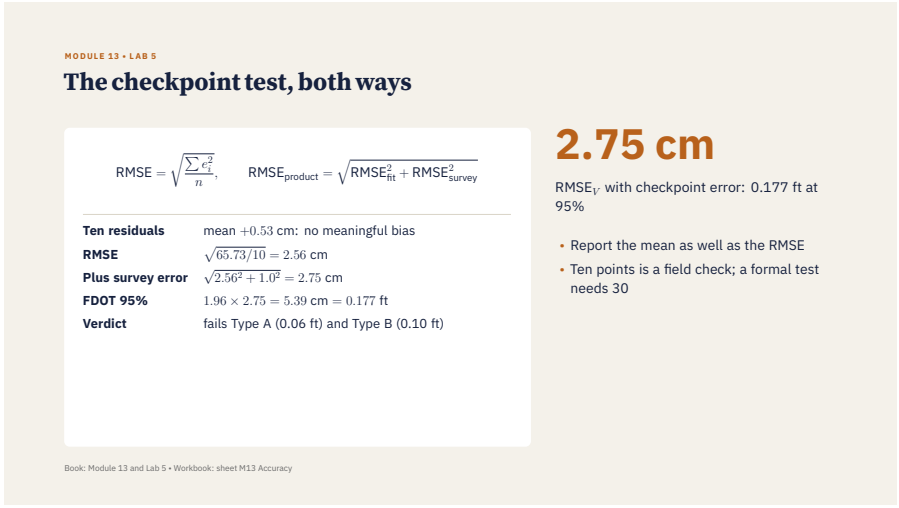
	ASPRS Edition 2 (2024)	FDOT Handbook (2025), TML
Statistic	RMSE, with the mean error stated	95% confidence (NSSDA)
Checkpoints	at least 30, up to 120	at least 20
Checkpoint accuracy	at least 2× better than the product	the PSM's choice (Note 5)
Checkpoint error	included in the result	not required
Relative accuracy	numeric limits within and between swaths	20% overlap
Vegetation	VVA reported as found	not addressed

Book: Modules 13 and 19 • Supplementary: FDOT Handbook Sections 39.6.2 and 39.7

The two standards a Florida surveyor meets on mobile LiDAR work, side by side. ASPRS Edition 2 reports RMSE from at least 30 checkpoints, twice as accurate as the product, with the checkpoint error included. FDOT tests at 95 percent under NSSDA with at least 20 checkpoints and leaves checkpoint accuracy to the surveyor. A dataset can pass one and fail the other, so the scope has to say which governs.



Leave this slide up. Divide the 95 percent tolerance by 1.96 for vertical and 1.7308 for horizontal. Type A needs an RMSE of 0.93 cm vertical and 1.06 cm horizontal. Type B needs 1.56 and 1.76. Compare with this morning: the wide-beam sensor's range bias alone was 1.25 cm, and GNSS vertical is 1 to 2 cm. Type A is at the edge of what mobile LiDAR can prove, and it is reached only by adjusting to dense control.



Lab 5, done together in the workbook. Ten vertical residuals give an RMSE of 2.56 cm. Add the checkpoint survey's own 1.0 cm in quadrature: 2.75 cm. That is the ASPRS number. Multiply by 1.96 for the FDOT number: 5.39 cm, or 0.177 ft. It fails both Type A and Type B. Always report the mean error too: a bias hides inside an RMSE, and the 95 percent factor assumes there is none.

MODULE 21 • THE CAPSTONE

Passes its ASPRS agreement. Meets neither FDOT type.

Item	Sample project	Type A	Type B
Vertical at 95%	$1.96 \times 2.06 = 4.04$ cm (0.132 ft)	≤ 0.06 ft: **fails**	≤ 0.10 ft: **fails**
Horizontal at 95%	$1.7308 \times 2.82 = 4.88$ cm (0.160 ft)	≤ 0.06 ft: **fails**	≤ 0.10 ft: **fails**
Point density	58 points/ft^2	≥ 20: meets	≥ 10: meets
Checkpoints	30, about 220 ft apart	meets	meets
GNSS	300 m outage, about 85% fixed	document (Note 4)	document (Note 4)

Book: Module 21 • Workbook: sheets M21 Capstone and Dataset

The book’s capstone is a resurfacing design survey, which is Type A work. It passes its ASPRS 3 cm vertical and 5 cm horizontal classes with room to spare. Converted to FDOT terms it is 0.132 ft vertical and 0.160 ft horizontal: over both the Type A and the Type B limits. Density and checkpoint spacing are fine. Reaching Type A would mean roughly halving the RMSE. Settle which standard governs before mobilization, not after the test.

MODULE 13 • RELATIVE ACCURACY

ASPRS data internal precision limits (cm)

Vertical accuracy class	Within swath	Between swaths, RMSD	Between swaths, max
Factor of the class X	$0.60X$	$0.80X$	$1.60X$
1 cm (near Type A)	0.6	0.8	1.6
1.5 cm (near Type B)	0.9	1.2	2.4
3 cm (the capstone)	1.8	2.4	4.8
5 cm (a Type C choice)	3.0	4.0	8.0

Book: Module 13, ASPRS Edition 2 Version 2, Table 7.2

FDOT asks for 20 percent overlap between passes but sets no number for how well they must agree. ASPRS does. For a vertical class X, smooth-surface repeatability within a swath is 0.60 X, the swath-to-swath RMS difference is 0.80 X, and the maximum is 1.60 X. Test this on every project by differencing opposing passes: it reveals trajectory problems that checkpoints between them can miss.

RECOMMENDATIONS DOCUMENT, PROBLEM 5

Control spacing for Type A: by conditions, not one number

Conditions	Maximum control spacing
FDOT recommendation, any conditions	1000 ft
Open sky, clean fixed solution throughout	500 ft
Tree canopy, buildings, intermittent fix	300 ft
GNSS outage (overpass, tunnel)	150–300 ft, with at least one point inside

Supplementary: FDOT mobile LiDAR standards, problems and recommendations

Type A leaves about 6 mm for trajectory error between control points, once checkpoint error and laser noise are taken out. The adjustment pins the cloud only at each control point. At 13 m/s, 1000 ft is 23 seconds between control points; 300 ft is 7 seconds, short enough that the error is nearly constant and the adjustment removes it. These spacings are the course's recommendation from the error budget, not a published standard. Confirm them with midpoint checkpoints on your own system.

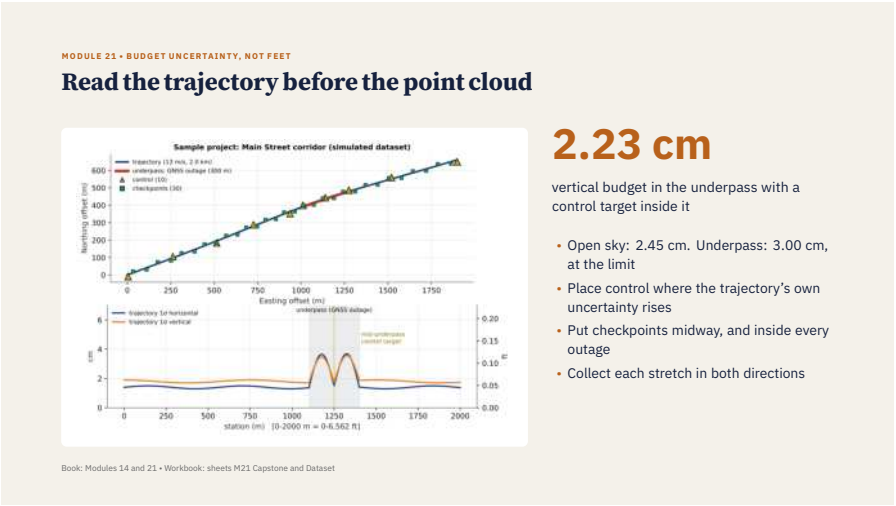
RECOMMENDATIONS DOCUMENT, PROBLEM 5

Scans between control points behave like traverse legs

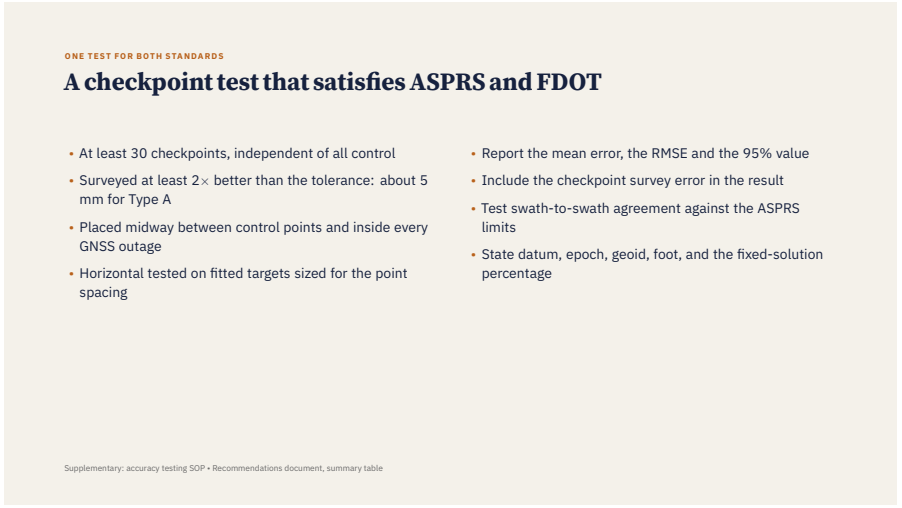
Speed	300 ft spacing	1000 ft spacing	Peak at the midpoint
13 m/s (29 mph)	about 70 scans, 7 s	about 235 scans, 23 s	about 4 mm against 8 mm
5 m/s (11 mph)	about 180 scans, 18 s	about 610 scans, 61 s	more scans, more drift time

Assumes 1 mm of independent relative error per scan, for illustration • 10 Hz scanner

Distance in feet is a stand-in. Between two control points the scans behave like the legs of a traverse: small independent errors accumulate with the square root of the number of scans, and the uncertainty peaks midway. At an assumed 1 mm per scan, 70 scans peak near 4 mm and 235 scans near 8 mm, against the 6 mm available. Slowing down adds scans and overlap but also adds drift time. Canopy, speed, turns and featureless scenes each weaken a different weight.



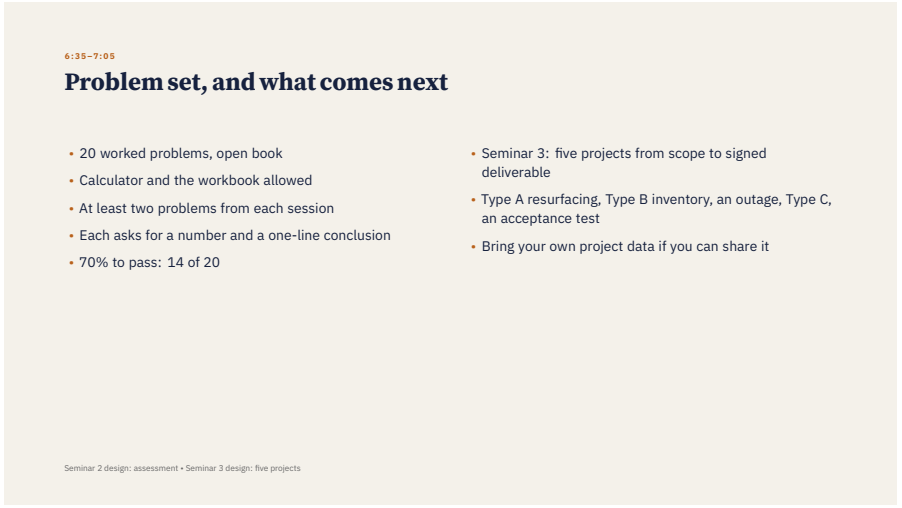
The trajectory's estimated standard deviation shows where accuracy is strong and where it is weak. In the capstone the vertical budget is 2.45 cm in open sky, 3.00 cm in the underpass, and 2.23 cm once a control target is placed inside the outage. That one target is what makes the job pass. Do not trust the reported uncertainty blindly: it is only as good as the software's noise models, and midpoint checkpoints are the proof.



One test done properly satisfies both standards. Thirty checkpoints covers FDOT's twenty. Surveying them twice as accurately as the tolerance means about 5 mm for Type A, which is leveling and total station work, not RTK. Report the mean, the RMSE and the 95 percent value together. Then the accuracy statement can cite ASPRS Edition 2 and the FDOT type in the same paragraph.



The takeaway of the day. Every session led here: what one return is worth, how the pose errors multiply, how the trajectory is adjusted, how the errors add up, what the hardware contributes, and finally what the standards demand and how to design a survey that passes them. The sensors keep improving; what remains is calibration, control and reference, which is the surveyor's part.



Thirty minutes for the problem set. Each problem asks for a number and a conclusion, such as meets Type B or outage too long for this IMU, so the grade tests judgment as well as arithmetic. Hand out the evaluation form with the problem set. Close by pointing to Seminar 3, which is projects only, and invite participants who can share de-identified project data to get in touch.